



Multi-modal fake news detection: A comprehensive survey on deep learning technology, advances, and challenges

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Abstract

The rapid development of social media and networks has dramatically accelerated the spread of multimodal fake news, posing significant threats to social stability, the public's right to information, and the credibility of news organizations. News modalities have advanced from simple text or images to combined video and audio formats. Integrating these heterogeneous data streams poses greater challenges for fake news detection. In recent years, the rapid advancement of Artificial Intelligence (AI) technology, including Large Language Models (LLMs), social network analysis, and external knowledge integration, has provided new insights and methods for detecting fake news. Nevertheless, there is a clear need for a thorough survey of multi-modal fake news detection. To address this, we have undertaken a comprehensive review to complement and consolidate existing research over the past five years. We systematically organize the key technologies and methods in this field and analyze the advantages of different technical approaches. Subsequently, we summarize mainstream fake news detection datasets and evaluation metrics, and analyze existing research directions. We discuss the challenges in this field, such as insufficient data annotation and model interpretability issues. Through this survey, we aim to provide researchers with a comprehensive understanding of fake news detection technologies and promote further development in this field.

Keywords Fake news detection · Deep learning · Multi-modal · LLM · Social media

1 Introduction

With the rapid advancement of internet technologies and social media platforms, individuals increasingly rely on digital communication and online information dissemination (Nadeem et al. 2023a; Raza et al. 2025). However, with the acceleration of information propagation, the spread of fake news has become increasingly rampant (Xu et al. 2023a). Fake news misleads the public and poses significant threats to social order, political stability, and public health (Ding et al. 2024). For instance, during the COVID-19 pandemic, much misinformation about the epidemic led to public panic and misunderstanding (Huang et al. 2020). Furthermore, multimodal fake news encompasses diverse formats, including text, images, videos, and audio. This heterogeneity poses significant new challenges, as solutions designed for unimodal data are often ill-suited for multimodal environments. Therefore, how to effectively detect and identify fake news has become a hot issue in both academia and industry.

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In recent years, the rapid development of Deep Learning (DL) technologies and their remarkable successes in fields such as image recognition and Natural Language Processing (NLP) have provided new insights and methods for fake news detection (Tufchi et al. 2023). DL models can automatically learn complex feature representations from data, offering unique advantages in the feature expression of multi-modal information in fake news, such as text, images, and videos (Ayetiran and Özgöbek 2024). For example, Convolutional Neural Networks (CNNs) can be employed to extract visual features from images (Segura-Bedmar and Alonso-Bartolome 2022), while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) are suitable for analyzing the temporal features of textual content (Cui et al. 2025). Graph Convolutional Networks (GCNs) can also be utilized to model user relationships and information propagation paths on social media (Zhou et al. 2024). In recent times, the emergence of new methods based on Transformer architectures (Wang et al. 2023a), Large Language Models (LLMs) (Wu et al. 2024), generative models (Sharma et al. 2024), ensemble learning (LekshmiAmmal and Madasamy 2025a), distillation mode (Xu et al. 2023a), and model fusion has further expanded the research horizons in the field of fake news detection. These advancements have enhanced detection accuracy and provided more robust solutions to the challenges posed by the complex and diverse nature of fake news.

Despite the significant progress achieved by DL-based approaches for fake news detection, numerous challenges and limitations remain. Early research on fake news detection focused predominantly on unimodal data, particularly text or image (Kishwar and Zafar 2021; Abdali et al. 2024). These studies cannot meet the detection needs of multimodal fake news data. Recently, multi-modal fake news detection, which integrates multiple modalities such as text, images, and videos, offers distinct advantages over single-modality detection by comprehensively capturing the complexity of false information. However, current research faces several critical challenges (Hamed et al. 2023). First, establishing effective feature fusion mechanisms across different modalities remains a key issue (Qu et al. 2024). It involves designing frameworks that can appropriately combine and weigh the intrinsic features of each modality to improve detection accuracy. Second, addressing the semantic gaps within and between modalities is crucial, as these gaps can lead to inconsistencies in feature representations (Wang et al. 2024a). Lastly, training efficient and scalable models is essential, especially given the increasing volume and complexity of such data. These challenges highlight the need for innovative solutions to enhance the robustness and generalizability of

fake news detection models in real-world scenarios (Fendt et al. 2023).

Several studies have comprehensively reviewed research methods in fake news detection, categorizing and exploring them from various perspectives (Liu et al. 2024a). Kondamudi et al. (Kondamudi et al. 2023) reviewed the background, attributes, and creators' intentions of fake news. Tufchi et al. (Tufchi et al. 2023) meticulously analyzed the life cycle of misinformation, from its origination and dissemination to detection. Additionally, there have been articles summarizing multi-modal fake news detection (Kishwar and Zafar 2021; D'Ulizia et al. 2021), reviewing popular fake news detection datasets, and conducting feature and comparative analyses of each dataset. Other studies have focused on research related to fake news detection models (Liu et al. 2024a; Zhou and Zafarani 2018; Bondielli and Marcelloni 2019; Alam et al. 2021; Oshikawa et al. 2020; Al-Tai et al. 2023; Li et al. 2025). However, most existing reviews discuss outdated methods and lack interpretations of the latest research findings and model frameworks in misinformation detection. Table 1 briefly overviews the existing survey papers and our research contributions.

To advance the development of the field of fake news detection, this paper aims to provide a comprehensive review and categorization of methods for multimodal fake news detection, with a particular focus on new approaches developed over the past five years.

Overall, the contributions of this survey can be summarized as follows:

- A technical overview and analysis of recent developments in multi-modal fake news detection, with an in-depth exploration of various methods' underlying principles, strengths, and weaknesses.
- We have systematically summarized and categorized multi-modal datasets, detailing their data composition, key characteristics, and sources for download. Furthermore, we have evaluated and compared the performance of various methods on these datasets.
- This review not only synthesizes the existing achievements but also identifies several promising research directions for multi-modal fake news detection. From this perspective, this article serves to guide potential researchers in identifying research gaps and to inspire engineers to develop advanced application models.

In summary, fake news detection based on multi-modal is a research area of significant importance and broad prospects. This paper reviews the relevant research progress over the past five years, analyzes the existing challenges, and explores future directions to provide valuable insights and guidance for developing fake news detection technologies.

papers on multimodal fake news detection published in major journals since 2018, as illustrated in Fig. 2. We can observe a consistent year-over-year publication volume since 2018, with particularly high publication frequencies maintained from 2021 onwards. This trend demonstrates that fake news detection has become a sustained research hotspot that attracts significant academic attention.

2.2 Methodology overview

According to temporal progression, we have organized the main models and methodological evolution in the multimodal fake news detection research field, as shown in Fig. 3. It can be observed that the field has evolved from initial simple fusion methods based on multimodal features to advanced deep fusion strategies incorporating social context modeling and knowledge enhancement technologies. Over

Fig. 2 The number of papers for fake news detection since 2018

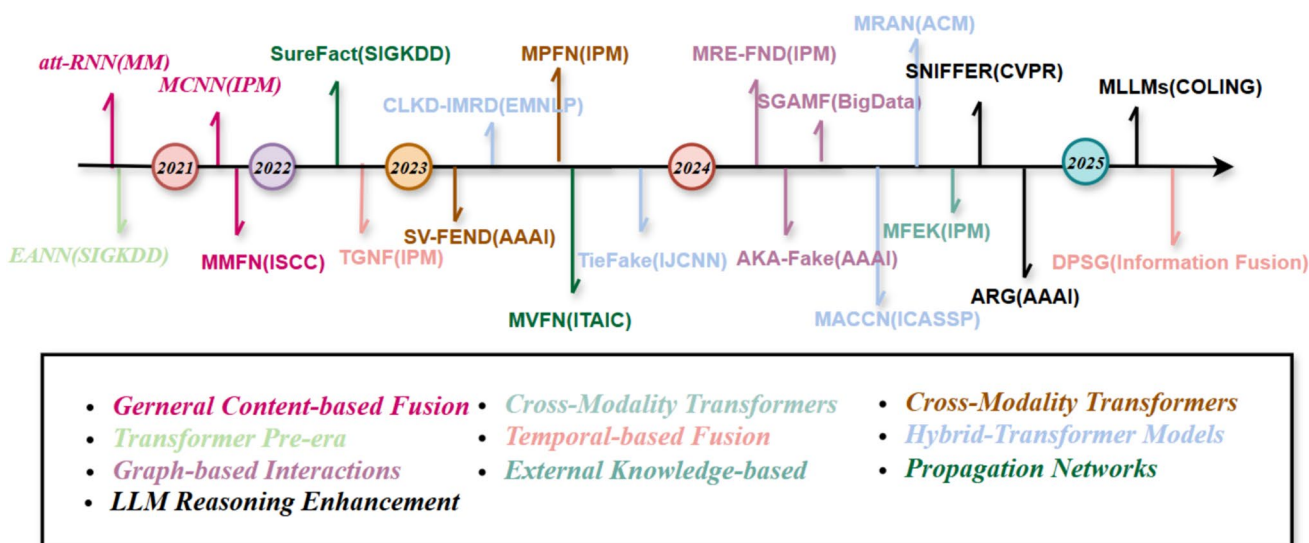
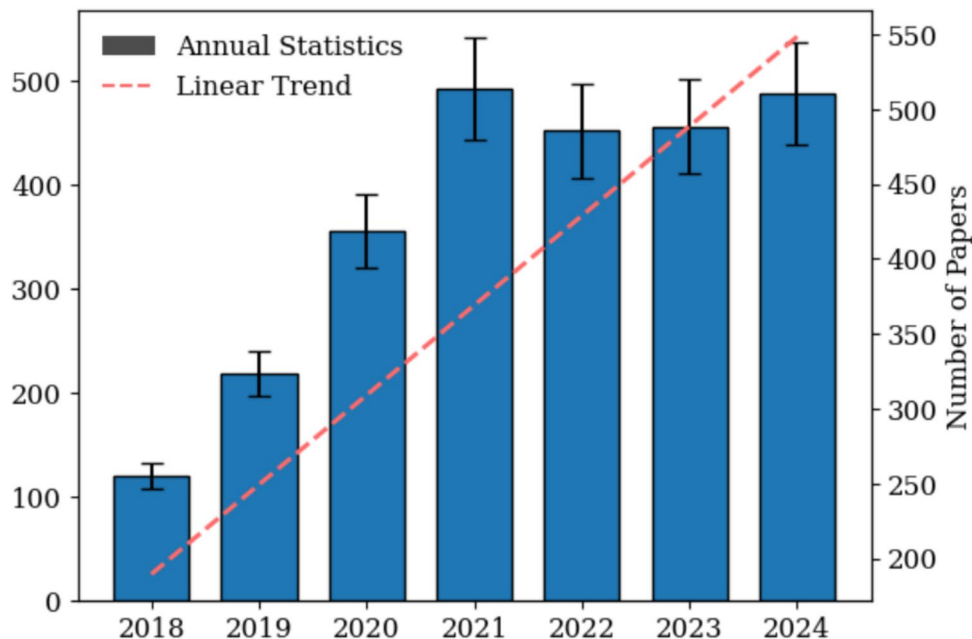


Fig. 3 The development chart of various methods is displayed along the time axis. One color represents a category, and typical models in each category are displayed on the timeline, with corresponding classification colors labeled

the past three years, GNN-based methods, social propagation modeling approaches, and graph-based interaction mining methods have developed in parallel, demonstrating outstanding performance in various dimensions. Meanwhile, Transformer architecture-based models and knowledge enhancement techniques have emerged prominently in fake news detection, significantly improving detection performance. More importantly, the rise of LLM reasoning technology marks a significant shift from traditional pattern recognition toward intelligent reasoning, opening new development pathways for building more intelligent and explainable detection systems.

Over the past five years, multimodal fake news detection methods have undergone substantial technological evolution. The initial phase was primarily defined using general content fusion techniques and elementary graph interaction models before 2021. Seminal works such as the Event Adversarial Neural Network (EANN) (Wang et al. 2018) laid the groundwork for multi-modal adversarial learning, while the attention-based RNN (Jin et al. 2017) introduced attention mechanisms for cross-modal feature integration. These early approaches primarily concentrated on basic feature extraction and shallow modal interactions.

In 2022, a paradigm shift emerged towards understanding information propagation patterns through a propagation network. Significant contributions include SureFact (Yang et al. 2022), which introduced a news detection pipeline integrated with social network analysis, and temporal-based fusion frameworks incorporating temporal dynamics into fake news detection. In 2023, the widespread adoption of the Transformer architecture fundamentally changed the landscape of multi-modal fake news detection. Cross-modal Transformers became the dominant technology, with models such as multi-modal pre-trained fusion networks demonstrating outstanding performance in cross-modal attention mechanisms, while the self-supervised visual-text feature enhancement model SV-FEND (Qi et al. 2023) introduced self-supervised learning techniques to improve feature representation. Meanwhile, hybrid Transformer models gained prominence, represented by cross-modal knowledge distillation learning model CLKD-IMRD (Xu et al. 2023b) for image-text multi-modal reasoning detection, TieFake (Guo et al. 2023), and MVFN (Cui et al. 2023), which integrated knowledge distillation techniques, temporal information enhancement, or multi-perspective analysis.

Based on the maturity of the Transformer architecture, a parallel development landscape has formed among entity-based methods, Social context modeling methods, and knowledge enhancement methods. Advanced graph neural architectures such as the multi-modal reasoning and attention network model MRAN (Yang et al. 2023), the multi-modal relation enhancement model MRE-FND (Han et al. 2024), and the social graph attention and multi-modal fusion

model SGAMF (Du et al. 2024) have demonstrated outstanding capabilities in modeling complex relational structures, while entity-based detection methods identify factual errors through deep mining of key entities and their semantic relationships, and Social Context Modeling methods analyze user behavior patterns and influence diffusion mechanisms from a communication perspective.

In the past two years, knowledge enhancement methods have reached unprecedented importance, with representative models such as SNIFFER (Qi et al. 2024), MFEK (Lin et al. 2024), and MACCN (Yi et al. 2024) significantly improving the credibility and interpretability of detection systems by integrating structured knowledge graphs, encyclopedia knowledge bases, and fact-checking databases. Moreover, the rapid development of LLM reasoning technology in recent two years has brought revolutionary changes to this field, with advanced systems such as multi-modal LLMs (Zheng et al. 2025) and dynamic prompt semantic generation model DPSG (Jing et al. 2025) fully utilizing the emergent reasoning capabilities of LLMs, capable of executing complex logical reasoning, commonsense reasoning, and causal reasoning, achieving a fundamental transformation from traditional pattern recognition to deep semantic understanding and reasoning judgment.

From the technical framework perspective, this paper statistically analyzes research methods on fake news using multimodal approaches over the past five years. We summarize and classify these researches, as shown in Fig. 4. These categories include: (1) General content based fusion model; (2) Attention mechanisms, which encompass both early and intermediate fusion techniques, as well as late fusion approaches such as CARMN (Song et al. 2021a); (3) Transformer-based models, including Transformer pre-era and cross-modality Transformers; (4) Graph-based interaction models, which leverage social network structures and temporal information; (5) Knowledge-enhanced strategies that integrate external knowledge based and LLMs such as SNIFFER (Xie et al. 2024) and OPEVFake (Wang et al. 2018). In the following sections, we will discuss them in detail.

For the advantages and disadvantages of each method, as well as applicable environments, we provide an overall overview, as shown in Table 2. We evaluate models from six aspects: (1) Computational Resource (CR) requirements evaluate the computational capabilities needed for model operation, directly affecting deployment feasibility; (2) Accuracy measures model performance, which served as the core effectiveness indicator; (3) Dataset Collection (DC) difficulty assesses the complexity of acquiring training data, impacting reproducibility; (4) Processing Speed (PS) measures response speed in practical applications, crucial for real-time scenarios; (5) Environment Adaptation (EA) analyze suitable application environments, helping researchers

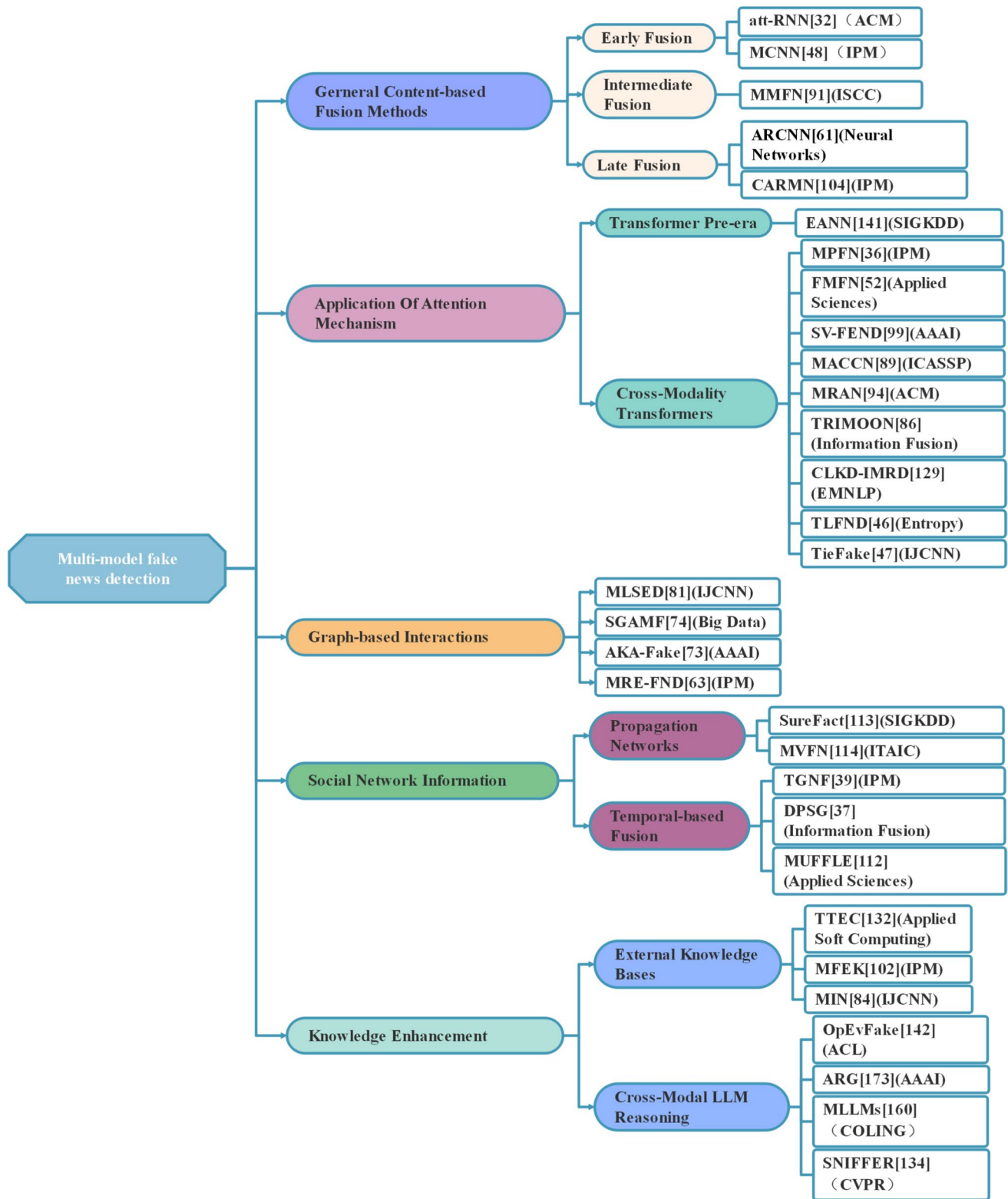


Fig. 4 Classification of research methods and representative papers

Table 2 Advantages, limitations, and applicable scope of multi-modal fake news detection architectures. (Computational Resource (CR): L (Large), M (Medium), S (Small), Accuracy (Acc): H(High), M(Medium), L(Low), S (Small), Dataset Collection (DC): L (Large), M (Medium), S (Small), Processing Speed (PS): F (Fast), S (Slow))

Methods		CR	Acc	DC	PS	EA	EK Dependency
General Content-based Fusion	Early fusion	S	L	M	F	Basic model, low computational requirements	N
	Cross-model fusion	M	H	M	F		N
	Late fusion	S	L	M	F		N
Application of the Attention Mechanism	Transformer pre-era	S	L	M	F	Attention mechanisms, low computational requirements	N
	Cross-modality Transformers	M	M	M	F	Pre-trained models, high-performance resources	N
	Hybrid-Transformer models	L	H	M	S	Transformer innovation, high computational	N
Graph-based Interactions		M	H	M	F	Entity linking, knowledge reasoning	N
Social Context Modeling	Propagation network	M	M	L	F	Graph model analysis, high data requirements	N
	Temporal-based fusion	M	M	L	F		N
Knowledge Enhancement	External knowledge-based	M	H	M	F	Frequency-time domain, external knowledge support	Y
	LLM reasoning enhancement	L	M	M	S	LLM reasoning, high computational	Y

choose appropriate solutions; (6) External Knowledge (EK) dependency evaluates reliance on external resources, affecting method independence and maintenance costs.

3 Methods based on multi-modal

We first describe the modality types and extraction methods in multi-modal fake news detection tasks in Section 3.1. In the subsequent sections, we systematically introduce different technical architectures for multi-modal fake news detection. Section 3.2 focuses on cross-modal content-based fusion, which includes early simple feature fusion models, the application of attention mechanisms, and graph-based interactions. Section 3.3 presents social context modeling, featuring two typical social network architectures: propagation network and temporal-based fusion. Finally, Section 3.4 covers knowledge enhancement, including external knowledge models and cross-modal LLM reasoning enhancement components.

3.1 Multi-modal data types and feature extraction

Data from each modality possesses its own characteristics. The purpose of multi-modal detection methods is to integrate multi-modal data and leverage complementary advantages, and it has been verified to significantly improve detection accuracy (Xue et al. 2021). The data used for fake news detection encompasses various types, including text, images, videos, and audio.

Feature extraction is the foundational step to isolate discriminative features from raw modalities. Effective feature extraction enables subsequent analysis modules to focus on the most relevant information for authenticity verification.

(1) **Text modality.** Text has long been a traditional medium for spreading misinformation and is one of the most easily disseminated forms of content in online communication. It consists of words, numbers, letters, and special symbols, thus forming news texts, user comments, etc. The content expressed through text is not only easily understood by humans but can also be easily interpreted by machines.

- **Text feature extraction.** Pre-trained language models, particularly Bert (Szczepański et al. 2021) and its variants (RoBERTa, ALBERT), have become the standard methods for extracting semantic representations from news text. These models capture contextualized word embeddings that preserve syntactic and semantic relationships. For instance, Szczepański et al. (Szczepański et al. 2021) utilized Bert to capture meaningful features that drive model behavior, while Singh et al. (Singh and Sharma 2022) employed RoBERTa for text processing in their multi-modal approach. LLMs such as Generative Pre-trained Transformer (GPT) have recently been used to obtain text features (Kakizaki et al. 2025).

(2) **Image modality.** As an intuitive form of visual information, images can provide evidence for fake news, making it appear more credible. For example, pairing authentic images with misleading headlines or descriptions can easily convince readers of the false content.

Images can be manipulated or selectively cropped to align with the fake news narrative. Combining real images with false text often creates subtle inconsistencies between the visual and textual content, and this inconsistency can be used to detect fake news.

(3) **Video modality.** Video data's multi-modal and temporal characteristics present challenges and new detection opportunities. Deepfake is currently the most prominent threat, where malicious actors can replace a person's facial expressions, lip movements, and actions with those of a target individual, fabricating statements or actions they never actually made.

- **Visual (image and video) feature extraction.** CNNs always dominate visual feature extraction, with architectures like VGG-19 (Guo et al. 2023; Wang et al. 2023b, 2022a), ResNet (Xue et al. 2021; Zhang et al. 2024a), EfficientNet-B0 (Singh and Sharma 2022), Vision pre-training models such as CLIP (Zheng et al. 2025; Yi et al. 2025), BLIP (Hetzner et al. 2025), BLIP-2 (Huang et al. 2023), InstructBLIP, and GPT-4 V (Nan et al. 2024). These networks capture hierarchical visual patterns that may reveal manipulation artifacts or inconsistencies. Recent approaches have incorporated Vision Transformers (ViT) for improved long-range dependency modeling. For example, Singh et al. (Singh and Kumar 2024) leveraged ViT to model long-range dependencies in visual data, while Sun et al. (Sun et al. 2023a) employed ViT for image feature extraction in their adaptive gradient graph interaction network. Video processing typically involves 3D-CNNs (Raza and Malik 2023) or separate processing of visual and audio streams.

(4) **Audio modality.** The audio portion of fake news videos often carries stronger emotional tones, while the emotional expression in real news videos tends to be more neutral and rational.

- **Audio feature extraction.** For audio modalities, 1D-CNNs and ResNet-1D architectures (Raza and Malik 2023) extract temporal acoustic features that may reveal manipulation artifacts.

(5) **Social network.** A social network graph provides key information, such as dissemination dynamics, user relationships, and group behaviors, far exceeding the news content. It is structurally defined by nodes and edges, which collectively model the interactions within information dissemination systems. The nodes may include user nodes, content nodes, IP nodes, and so on. The edges include social edges, propagation edges, content edges, and so on. Researchers used social relationships (Shu et al. 2018) and heterogeneous graph information (Dou et al. 2021) to detect fake news.

- **Social network feature extraction.** The most mainstream method currently is GNNs, which automatically learn low-dimensional vector embeddings of nodes in a graph (Yuan et al. 2021), thereby encoding

the structure and positional information of the nodes. Jing et al. (Jing et al. 2025) used dynamic propagation graphs to depict the temporal sequence of news dissemination.

- **Cross-domain feature alignment.** Some methods focus on aligning features across modalities to enable more effective fusion. Li et al. (Li et al. 2021) proposed an entity-oriented multi-modal alignment approach where textual entity features are extracted using Bert, while images are processed through Faster RCNN to identify and extract entity features. Similarly, Zhong et al. (Zhong et al. 2023) extracted fine-grained news features from salient regions in images and semantic words in text, employing a cross-modal weighted residual network to learn shared semantic clues.

3.2 Cross-modal content-based fusion method

According to the multi-modal fake news detection architecture that has been summarized in recent years, content-based methods are one of the most common approaches. First, feature extraction models are employed to extract features from multimodal data, including text, images, audio, video, and network information. Second, attention mechanisms, distillation, and fusion are utilized to refine, align, and integrate multi-modal features at a fine-grained level. Some studies leverage specific features from social media as detection criteria. The characteristics of communication dynamics, user behavior, network topology, and other information can identify fake news. The typical news detection methods based on text and image modalities are shown in Fig. 5.

In this section, we categorize detection methods based on multi-modal data content into three subsections according to their primary technical foundations: (1) general content-based fusion methods; (2) application of attention mechanism; (3) Graph-based interactions. General content-based fusion methods play a crucial role in early model design, particularly through feature extraction and fusion mechanisms, thereby closely connecting content-oriented research with feature-oriented research. These methods primarily include three strategies: early fusion, late fusion, and cross-model fusion, providing a fundamental framework for integrating information from different modalities. Attention mechanism-based methods, with the introduction of the Transformer as a watershed moment, demonstrate significant differences between the Transformer pre-era and the Transformer post-era, fundamentally transforming the processing of sequential data and multi-modal information. Graph-based interaction methods achieve a paradigm shift from traditional feature-level fusion to semantic-level understanding, constructing cross-modal associations at the semantic level by identifying and aligning multi-modal entity information.

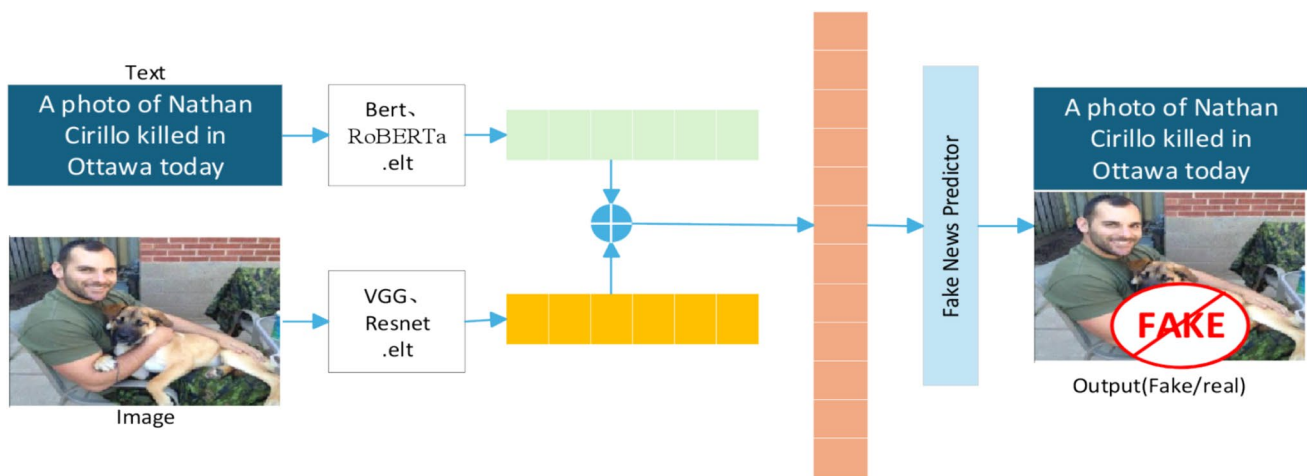


Fig. 5 Example of multi-modal fake news detection methods based on content features (Taking text and image as examples)

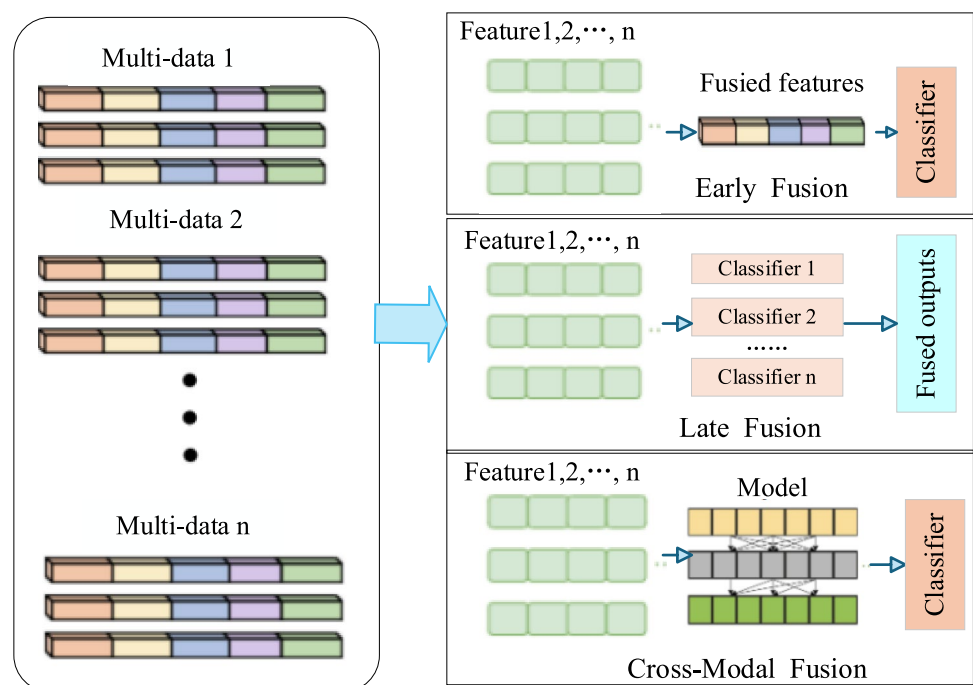
3.2.1 General content-based fusion

Feature extraction and fusion mechanisms are crucial in model design (Qu et al. 2024). According to content-oriented research, multimodal data features are fused through different fusion methods. We divide this part into three types: (1) Early fusion (Qu et al. 2024; Xue et al. 2021); (2) Late fusion (Wang et al. 2022b; Kananian et al. 2023); (3) Cross-model fusion (Guo et al. 2023; Wu et al. 2023a), as shown in Fig. 6.

Early fusion The early fusion method involves integrating features from multiple modalities at an initial stage, typically

through operations such as concatenation. Such fusion is commonly conducted before the classification step (Raj and Meel 2022). When the integration occurs after feature extraction, it may be termed intermediate fusion (Agarwal et al. 2019). Due to the overly simplistic feature extraction approach in early fusion, many scholars have optimized this method at the model level (Wang et al. 2023b; Zhang et al. 2024a; Nakamura et al. 2019). Wang et al. (Wang et al. 2023b) improved fusion effectiveness by adopting a two-level text feature extraction module, an image extraction and fusion module, a three-level feature matching scoring module, and a multi-modal integrated recognition module,

Fig. 6 Categories of the general content-based fusion methods can be divided into three types: early fusion, late fusion and cross-modal fusion



while introducing Chebyshev distance metrics to calculate cross-modal matching scores. Similarly, Zhang et al. (Zhang et al. 2024a) built upon early fusion by utilizing ResNet-50 to leverage the complementarity of cross-modal features, fusing semantic, visual, and text.

Late fusion Late fusion methods integrate decisions from outputs of different modal models (Abdali et al. 2024). Common approaches in this category include majority voting, fuzzy decision fusion, average fusion, and LF-LSTM (Boulahia et al. 2021). Decision-based methods typically ignore contextual information, where powerful modalities with large sample sizes or high mutual information between features and labels may mislead decisions, and fail to provide space for weaker modalities to express themselves (Raj and Meel 2022; Chen et al. 2022). For example, Raj et al. (Raj and Meel 2022) introduced the ARCNN framework by combining RNN and CNN, and through studying early or late fusion methods, concluded that late fusion can achieve higher accuracy.

Cross-model fusion Cross-model fusion has successfully utilized state-of-the-art fake news detection (Guo et al. 2023). It was initially embodied in recurrent or convolutional deep models (Dong et al. 2023), eventually evolving into attention-based methods (Han et al. 2024; Sun et al. 2022). Recently, most research has been based on this structure (Du et al. 2024; Li et al. 2021; Wang et al. 2024b; Yang et al. 2021). In this subsection, we discuss early research of this type, and subsequent chapters will provide detailed discussions based on different technical characteristics.

Considering the inevitable information gap between different modalities, the main problem that cross-model fusion aims to solve is cross-modal alignment components to map different features into a common semantic space (Dong et al. 2023). Based on different approaches to feature mapping processing, scholars have designed various mapping methods, such as multi-path segmentation (Guo et al. 2023), entity matching (Li et al. 2021), temporal fusion units (Sun et al. 2022), etc. Considering the consistency of multimodal data, Guo et al. (Guo et al. 2023) proposed a model based on the ResNet architecture, capturing local and global

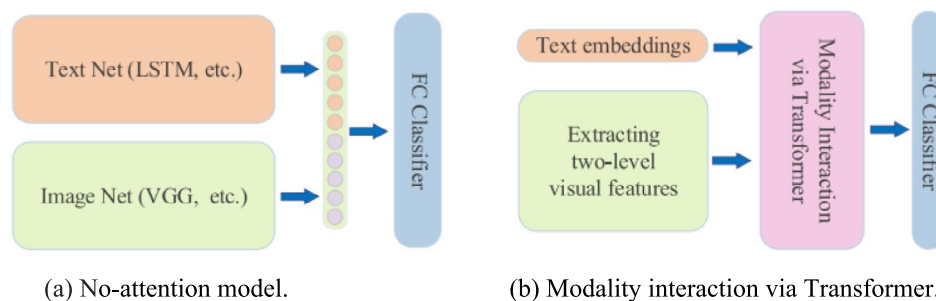
features through multi-path segmentation. Wu et al. (Wu et al. 2023a) proposed a multi-reading habit fusion inference network. Based on fine-grained entity analysis, Li et al. (Li et al. 2021) proposed an entity-oriented multi-modal alignment and fusion network structure for fine-grained entity analysis. Regarding innovation in CNN combined with a multi-modal fusion architecture, Dong et al. (Dong et al. 2023) combined DL and Markov random fields to model multi-modal dependencies. On this basis, to achieve a transparent decision-making process, Han et al. (Han et al. 2024) proposed an interpretable heterogeneous network inference framework. Leveraging the architecture of the temporal fusion unit, Sun et al. (Sun et al. 2022) developed a dual dynamic GCN to capture information propagation dynamics and contextual information from knowledge graphs.

3.2.2 Application of attention mechanism

Attention mechanism, as a core technology in the field of DL, dramatically enhances the ability of models to capture long-range dependencies and key information, and is widely used in the field of multi-modal fake news detection (Szczepeński et al. 2021; Singh and Sharma 2022; Wu et al. 2023a; Zeng et al. 2024; Guo et al. 2024; Wang et al. 2023c). Early approaches combined information from different modalities through simple feature concatenation or fusion, which could not distinguish between key and redundant information (Wang et al. 2018; Jin et al. 2017). Transformer enables a deep modality interaction process, which has made progress in capturing the correlation of multi-modal features more accurately and improving classification performance (Moattari and Zulkernine 2024), as shown in Fig. 7. It has currently achieved state-of-the-art performance in fake news detection (Qian et al. 2022; Wu et al. 2023b, c; Peng et al. 2024; Su et al. 2022; Hu et al. 2024a; Mu et al. 2023). We divide this research according to Transformer technology into two types: (1) Transformer pre-era; (2) Cross-Modality Transformers.

Transformer Pre-era Before the advent of Transformer architecture, attention mechanisms in multi-modal fake news

Fig. 7 The left model only splices features, and the right side shows that the Transformer realizes deep cross-modal interaction (Wang et al. 2023c)



detection were primarily based on traditional neural network architectures (Gu et al. 2019; Delbrouck and Dupont 2017). Representative work of this period is att-RNN (Jin et al. 2017). Jin et al. (Jin et al. 2017) proposed a neural-level attention mechanism based on LSTM, which enables the model to dynamically focus on key words and phrases in text sequences by introducing attention weight allocation on the hidden states of RNNs. This method achieved significant progress in identifying linguistic patterns of fake news.

The attention mechanism can identify important local features within each modality (Agarwal et al. 2019; Li et al. 2024a; Yan et al. 2024). Li et al. (Li et al. 2024a) used attention to point to the field of multi-modal deep forgery. The method enhanced the semantic representation of each mode through cross-modal knowledge injection. Yan et al. (Yan et al. 2024) proposed a one-dimensional attention mechanism module to process the input text and image sequence, thereby enhancing the essential aspects of the bimodal features.

After concatenating the representations of different modalities into a single sequence, the attention mechanism can automatically learn the relationships between cross-modal elements (Wang et al. 2024b; Meel and Vishwakarma 2023; Chen et al. 2021). For instance, Meel et al. (Meel and Vishwakarma 2023) integrated the transfer learning method into the attention mechanism to optimize the text and image features. In order to align visual evidence with text at granular levels, Wang et al. (Wang et al. 2024b) addressed multi-modal misalignment using cross-modal attention.

Some methods increasingly adopt dynamic attention to contextually adjust feature importance (Du et al. 2024; Zhang et al. 2024b). Zhang et al. (Zhang et al. 2024b) proposed a new adaptive knowledge-aware fake news detection model, which generated an adaptive knowledge sub-graph based on news content and merged it into the news through heterogeneous graph learning. Du et al. (Du et al. 2024) proposed an improved Albert model as a text encoder and designed a sparse gated attention mechanism to filter irrelevant features to optimize multi-modal fusion. Pan et al. (Pan et al. 2024) used multi-stage attention to refine features at both local and global levels.

Cross-modality transformers The emergence of Transformer has provided an efficient parallel computing framework for multi-head attention. Based on this approach, most scholars have recently enhanced cross-modal interaction capabilities, including self-attention (Xu et al. 2023b; Guo et al. 2024; Liu et al. 2023; Li et al. 2024b) and co-attention (Yi et al. 2024; Qian et al. 2022; Yin et al. 2023), which is presented in Table 3. Due to the continuous development of attention mechanisms, there is no unified distinction, but fundamentally, they are all based on dot-product attention mechanisms (Wang et al. 2022b).

Self-attention (SA) enhances the extraction of modal information, which can solve problems such as gradient explosion (Singh and Sharma 2022). When SA is introduced, it is typically combined with other methods to enhance modal fusion capabilities, such as combining contrastive learning (Wu et al. 2023b) or stacking multiple SA layers (Peng et al. 2024; Wu et al. 2023c). For example, Wu et al. (Wu et al. 2023b) introduce contrastive SA, dividing instances within a batch into positive and negative samples. Similar to contrastive learning methods, to optimize modal fusion, Wang et al. (Wang et al. 2023c) used multi-head SA to perform multi-modal fusion by calculating similarity between any two modalities.

In Co-Attention (CA) based methods, there are roughly three approaches: (a) stacking CA architectures between different modalities; (b) adding modular tasks during the process, such as adding pre-training tasks (Hu et al. 2024a) or gated fusion modules (Xiong et al. 2023); (c) combining other multi-modal interaction methods, such as using entity features (Yin et al. 2023; Zhang et al. 2024e). Stacking multiple CA architectures for multi-modal data represents a typical early approach to implementing CA mechanisms. Yu et al. (Yu et al. 2023) used two identical CA modules, while Wu et al. (Wu et al. 2023a) used three identical modules. A common practice in CA is to introduce modular models to enhance the generalization ability of the model. In order to obtain fine-grained feature representations of each modality through various modal features, Yang et al. (Yang et al. 2023) proposed a relation-aware attention module, while Hu et al. (Hu et al. 2024a) added an ITM pre-training task to the CA process, using modal information to complete the pre-training task and enhance cross-modal interaction capabilities. Based on the guided attention method, Zou et al. (Zou et al. 2023) proposed a multi-level interaction network for multimodal rumor detection. The CA model adopted by this network consists of CA blocks, each of which contains two SA blocks and a guided attention block, as shown in Fig. 8. Xiong et al. (Xiong et al. 2023) fused CA blocks with a gate-based fusion module. With the help of the semantic flip method, Yi et al. (Yi et al. 2024) improved the cross-modal model by adding semantic flipping in the text modality before joint attention. For deep multimodal feature interaction, Xu et al. (Xu et al. 2023b) concatenate two enhanced features with the initial review features to obtain the final multimodal features.

3.2.3 Graph-based interactions

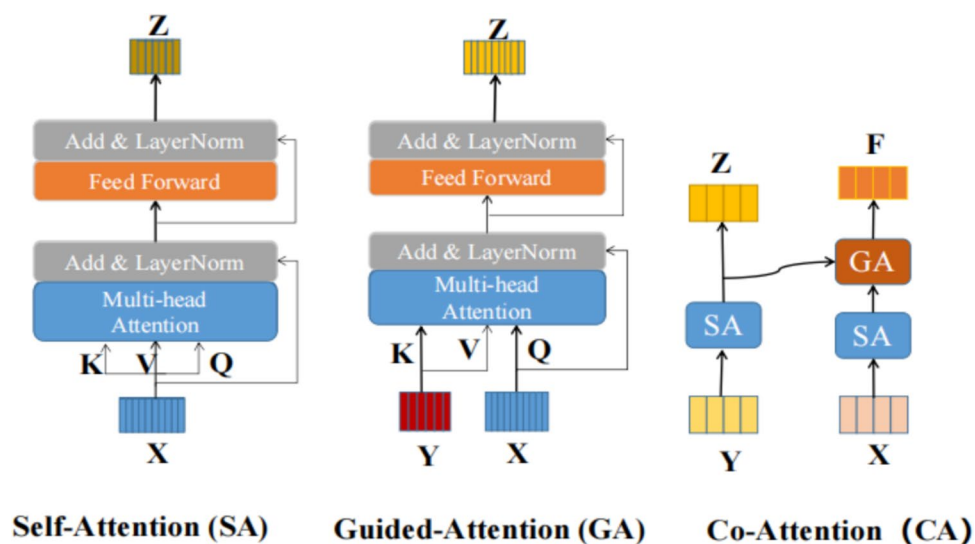
Compared to traditional feature-level fusion methods, Graph-based interaction methods shift the research focus from feature-level fusion to semantic-level understanding

Table 3 The summary of the application of Transformer-based models in multimodal fake news detection primarily includes information on methods, technology, suitability, and limitations. (SA-Trans: Self Attention (SA) Transformer; SA-CM: SA and cross-modal Trans-

former; S-Trans: Integration of sentence Transformer and SA; CL-A: Citing the SA in contrastive learning; S-A: Based only on SA; C-A: Integration of SA and cross-attention; GFM: Gate-based fusion module; RA: Relationship-aware; ITM: ITM-aware)

Ref	Year	Transformer Module		Technology	Suitability	Limitation
Singh and Sharma (2022)	2022	SA-Trans	SA-Trans	Image analysis and text comprehension	Solve problems such as gradient explosion	Insufficient capture of local information
Hu et al. (2023)	2023		CL-A	Multi-modal retrieval enhancement	Novel method	Large memory consumption
Wu et al. (2023b)	2023		S-A	Explainable semantic inconsistency reasoning	Strong interpretability	Difficult modal alignment
Wu et al. (2023c)	2023			Human cognition-inspired collaborative reasoning	Good effect	Large memory consumption
Wang et al. (2023c)	2023			Two-stage visual feature Transformer	Calculation of correlation	Greatly affected by various correlations
Zou et al. (2023)	2023	SA-CM	C-A	Three-layer attention-based hierarchical interaction	Two separate cooperative attentions to increase efficiency	Large memory consumption
Wu et al. (2023a)	2023			Reading habit-based consistency reasoning	Three co-attentions are more efficient	Large memory consumption
Yi et al. (2024)	2024			Adaptive cooperative attention	Add synonym replacement modules	Only for text modality
Xiong et al. (2023)	2024		GFM	Contrastive learning based two-round inconsistency	Add gating modules, with substantial flexibility	Poor real-time performance
Yang et al. (2023)	2024		RA	Relation-aware attention mechanism	Fine-grained feature representation	Large number of parameters
Hu et al. (2024a)	2024		ITM	Matching-aware cooperative attention mechanism	Add pre-training tasks with relatively high calculation efficiency	Sensitive to hyperparameters

Fig. 8 An example of Co-attention (CA) Network (Zou et al. 2023). SA and GA modules are general attention architectures, each incorporating a multi-headed attention and a fully connected network. The CA Network contains N stacked CA blocks, each of which integrates two SA modules and one GA module



by identifying and aligning multimodal entity data (Li et al. 2021; Pan et al. 2024). This type of method utilizes various entity information in multimodal data to establish cross-modal associations at the semantic level (Zhang et al. 2024a). More importantly, extracting information directly from entity data has good interpretability (Huang

et al. 2023). However, establishing a basic entity extraction and alignment framework is far from sufficient (Müller-Budack et al. 2021; Osman and Barukub 2020). Therefore, researchers have proposed targeted improvement plans from different perspectives, including improving entity matching mechanisms (Li et al. 2021), enhancing

the dynamics of knowledge graphs (Zhang et al. 2024b), optimizing entity type selection (Yin et al. 2023), processing specific types of entities (Yi et al. 2025; Müller-Budack et al. 2021), and introducing external knowledge validation (Han et al. 2024; Yin et al. 2023).

In response to the problem of insufficient accuracy in entity matching, Li et al. (Li et al. 2021) proposed an entity-oriented model that adopts an entity-level CA mechanism. By fusing entity attribute information, it achieves more accurate consistency judgment than simple entity name matching. To address the issue of static knowledge graphs being unable to adapt to dynamic news scenarios, Zhang et al. (Zhang et al. 2024b) proposed a reinforcement adaptive knowledge learning method that uses multimodal matching entities as seed nodes to adaptively explore relevant entity neighbors through reinforcement learning, generating compact and relevant knowledge subgraphs. The SceneFND model proposed by Zhang et al. (Zhang et al. 2024a) focuses on the distribution differences of three types of scene entities: location, weather, and season, to address their special role in false news detection. Due to the difficulty of accurately quantifying entity consistency using traditional similarity calculation methods, Müller-Budack et al. (Müller-Budack et al. 2021) introduced entity consistency scores to quantify the semantic matching degree between text entities and visual entities by constructing entity context correlation graphs.

In addition, some methods incorporate external knowledge sources, such as Wikipedia (Han et al. 2024) and BaiduAPI (Yin et al. 2023), which provide factual support for entity verification through entity linking and knowledge inference, thereby forming a comprehensive technical pipeline from entity extraction to knowledge enhancement (Zhang et al. 2024b). Considering the limitations of relying solely on news content for entity verification, the MRE-FND model proposed by Han et al. (Han et al. 2024) constructs a ternary association structure to verify the authenticity of news by linking news entities with Wikidata entities. Similarly, Yin et al. (Yin et al. 2023) focused on key descriptive objects in news images, extracting representative entities such as names, landmarks, specific organizational identifiers, strong visual impact concepts, general objects, and scenes, and implementing entity detection using the Baidu API and the Spacey open-source library.

3.3 Social context modeling method

Social context modeling methods by analyzing information propagation characteristics or user behavior on social media mainly include two core technical categories: propagation

network modeling (Zhang et al. 2024c; Zeng et al. 2022) and temporal-based fusion (Song et al. 2021b; Zeng et al. 2022).

Propagation network methods use GNNs to capture news propagation structures (Silva et al. 2024), achieving detection through constructing propagation graphs, extracting network features, and classification prediction (Jarrahi and Safari 2023). Temporal-based fusion methods focused on capturing temporal dependency features of fake news propagation and learned dynamic evolution patterns of the propagation process (Jing et al. 2025). However, there are some technical limitations; for example, propagation network methods suffer from simplified graph modeling and low computational efficiency, while temporal-based fusion methods are constrained by inadequate traditional static feature modeling (Lv et al. 2024). To address these challenges, researchers have improved graph construction strategy optimization (Yin et al. 2024), subgraph reasoning mechanisms (Yang et al. 2022), user behavior fusion (Wu et al. 2022), and continuous-time dynamic graphs (Song et al. 2021b), etc. A summary of the research on this method is in Table 4.

3.3.1 Propagation network

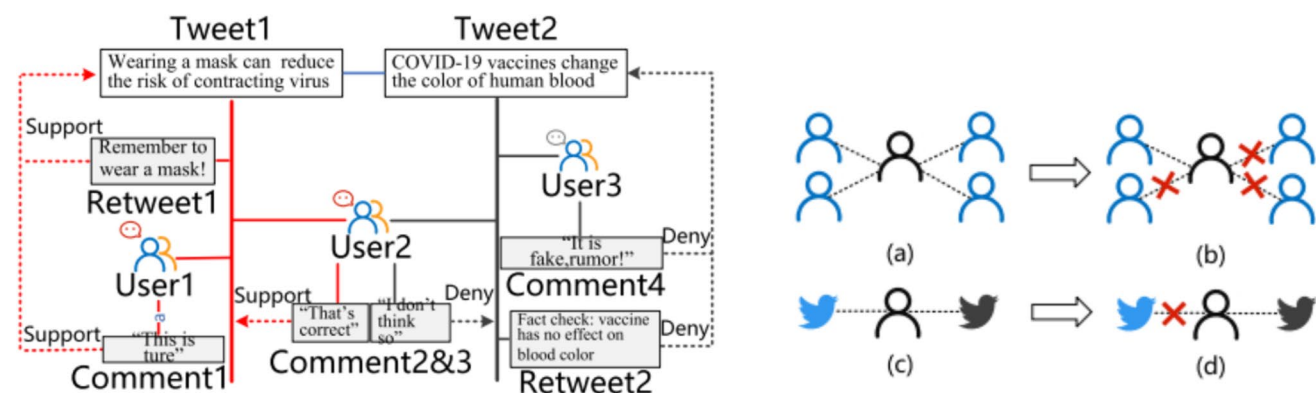
Information propagation paths, structural characteristics, and propagation patterns on social media help identify fake news (Wu et al. 2022). From the left of Fig. 9, we can observe that two initially unrelated tweets, Tweet1 and Tweet2, may become connected through the retweeting or commenting actions of the same user, such as User2 (Cui et al. 2023). Furthermore, the comments and opinions expressed by users in response to the tweets can, to some extent, reflect the credibility of the original content (Jing et al. 2025; Jarrahi and Safari 2023). However, inaccurate or irrelevant interactive connections within the propagation network may somewhat compromise the precision of rumor identification. Spurious interaction is displayed in the right of Fig. 9 (Liu et al. 2022). Fake and real news always exhibit different structural characteristics and diffusion patterns in the propagation network. Some studies utilize construction propagation network graph structures to improve the accuracy of fake news detection (Yang et al. 2022; Wu and Hooi 2023; Silva et al. 2021).

We summarize two common processing approaches: utilizing attention mechanisms to enhance the graph's anti-interference capability (Jing et al. 2025); and introducing unique modal alignment modules to strengthen multi-modal information interaction capabilities (Zhang et al. 2024d), such as adding CNN modules (Jarrahi and Safari 2023) and employing different adversarial neural networks for each domain (Zhou et al. 2024).

Some studies incorporated multi-head attention modules into social networks to enhance the anti-interference

Table 4 Summary of research on Social Context Modeling architecture (T: text; I: image; V: video; G: graph; A: audio)

Social context module		Ref	Year	Technology	Advantage	Limitation	Feature Extrac- tion
Propagation Network	Structural char- acteristics	Zhou et al. 2024)	2024	Contrastive learning	Good effect in the field	Sharp perfor- mance decline	T(BERT); I(CNN)
		Yang et al. 2022)	2022	Reward-punishment-based reinforcement learning	Good resistance to noise and interference	High computa- tional complex- ity	T(NLP); G(GNN)
		Cui et al. 2023)	2023	Attention-based multi-view fusion network	Good robust- ness	Large memory consumption	T(GRV); (CNN); G(GNN)
		Jarrahi and Safari 2023)	2023	User credibility assessment integration	Multi-scale fea- ture fusion	Large memory consumption	T(NLP); I(SLCNN)
		Yin et al. 2024)	2024	GAE with a concealment mechanism	Good resistance to noise and interference	Strong depend- ence on graph structure	T(NLP); G(GNN)
		Liu et al. 2022)	2022	Content-graph dual atten- tion mechanism	Good resistance to noise and interference	Strong depend- ence on graph structure	T(BERT); G(GCN)
		Wu and Hooi 2023)	2023	GNN-based social graph edge noise correction	Promotion of joint partici- pation	Lack of his- torical data for new users	T(NLP); G(GNN)
		Sun et al. 2023b)	2023	Cross-modal content-knowledge	Friendliness to parallel computing	Large memory consumption	T(BERT); G(GNN)
		Information propagation paths	Jing et al. 2025)	2025	Temporal fusion	Multi-scale fea- ture fusion	Insufficient emantic under- standing
	Propagation patterns	Zhang et al. 2024d)	2024	Diffusion-guided propaga- tion path generation	Multi-scale fea- ture fusion	High require- ment for labeled data	T(NLP); G(GNN)
Temporal- based Fusion	Co-attention (CA)	Jing et al. 2025)	2025	Temporal fusion	Grained time processing	Proneness to overfitting	T(BERT); I(CNN); (GCN)
		Wu et al. 2023b)	2021	Temporal memory module	Alleviation of the informa- tion sparsity	Low interpret- ability	T(BERT); I(ResNet-50)
	Fixed time block	Yin et al. 2024)	2022	GAE with a concealment mechanism	Multi-dimen- sional feature fusion	Interval size selection	T(NLP); G(GNN)

**Fig. 9** Left: Propagation of tweets. Right: Spurious interaction. **a** Fake followers. **b** Exclude interference. **c** Irrelevant tweets. **d** Exclude interference (Liu et al. 2022)

capability of social graphs (Jing et al. 2025; Wu et al. 2022; Liu et al. 2022). To address the difficulty in capturing

essential knowledge in social graphs, (Yang et al. 2022) proposed a reinforced subgraph reasoning module to capture

essential node data. Similar to it, (Wu and Hooi 2023) considered a pair of news articles connected by ordinary users and calculated common participation to reduce edge noise weights. Different subgraphs may have inconsistencies, so (Sun et al. 2023b) designed two consistency sub-networks specifically to capture information inconsistency across different modalities. To address data sparsity in early propagation stages, (Zhang et al. 2024d) utilized historical interaction information to generate propagation sequences. Similarly, (Silva et al. 2021) employed unsupervised learning and supervised domain-independent news classification modules, respectively.

Meanwhile, due to its inability to fully utilize dynamic interactions in a social network, optimization of role representations occurred. In this regard, (Jarrahi and Safari 2023) studied the effectiveness of publishers' core characteristics such as credibility, influence, sociality, validity, and lifetime.

3.3.2 Temporal-based fusion

Capturing the temporal dependencies between nodes in propagation patterns and the attention strength given to information from different temporal nodes is the main problem in temporal-based multi-modal fake news detection (Jing et al. 2025). Regarding how temporal information is integrated into static networks, we comprehensively analyzed temporal-based detection methods and summarized the following directions: (1) Using attention mechanisms

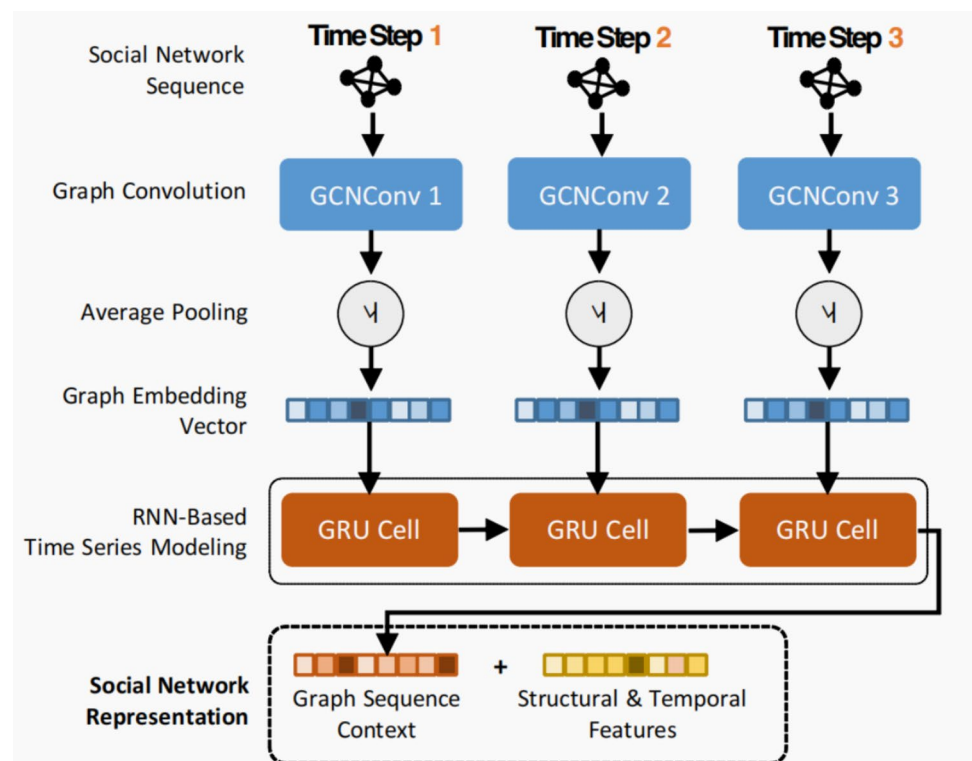
to give different attention weights to information from different temporal nodes (Jing et al. 2025; Song et al. 2021b); (2) Dividing different temporal nodes and assigning different weights to information from different temporal nodes, with weights being higher for nodes closer to publication time (Wu et al. 2022). In summary, temporal-based methods primarily focus on assigning different attention levels to various temporal nodes and dynamically adjusting the weights of multimodal information according to these attention levels.

For example, Song et al. (Song et al. 2021b) introduced temporal graph attention networks and set up a temporal memory module to store and update nodes. (Wu et al. 2022) combined the timeliness of news information to divide fixed observation windows while incorporating temporal statistics of users' historical tweets, as shown in the Fig. 10.

3.4 Knowledge enhancement method

Knowledge enhancement methods obtain detection capabilities by introducing additional knowledge beyond model training data (Li et al. 2025). External knowledge-based provides factual knowledge and structured information, while LLM reasoning enhancement provides logical reasoning and common knowledge. Together, they compensate for the shortcomings of traditional DL models that rely solely on training data pattern recognition while lacking world knowledge understanding and logical reasoning capabilities.

Fig. 10 Incorporating fixed step time blocks (Time Step 1, 2, 3...) into the model with different weights (Yin et al. 2024)



Therefore, we integrate external knowledge-based and LLMs' reasoning capabilities into enhancement methods. A summary of the research on this method is in Table 5. This section addresses the topic through two key dimensions.

3.4.1 External knowledge-based

Integrating external knowledge, such as structured knowledge graphs, domain-specific corpora, Wikipedia, LLMs, and web APIs, has significantly advanced the fake news detection field, as shown in Fig. 11. Researchers used this knowledge to retrieve, represent, and reason to overcome the inherent limitations of purely data-driven neural models (Wang et al. 2023d). Especially in the past two years, LLMs have been widely used in multi-modal fake news detection. We discuss the use of external knowledge outside of LLMs in this section, and will elaborate on the methods of LLMs in.

3.4.2 Subsection

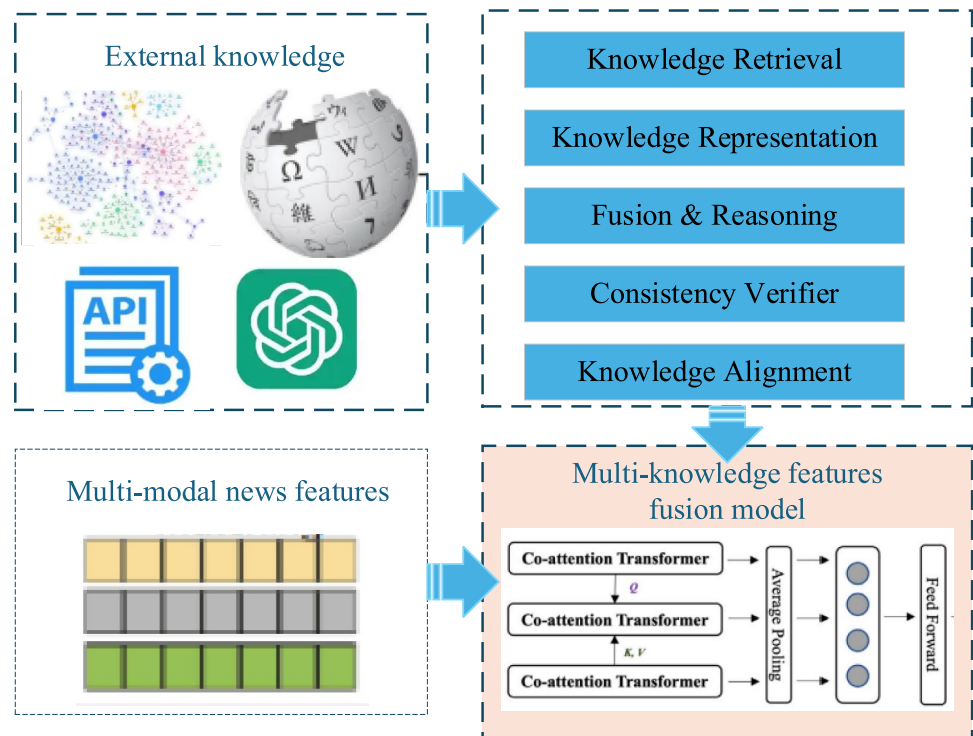
The external knowledge extraction component encompasses both explicit and implicit knowledge acquisition

from Wikipedia (Vrandečić and Krötzsch 2014), ChatGPT (Brown et al. 2020) or other structured knowledge (Vedula and Parthasarathy 2021). For instance, Zou et al. (Zou et al. 2023) proposed the MIN model, extracting entities from news text through the TAGME tool, images through YOLOv3 and Baidu image recognition API, and crawling descriptive external knowledge from Wikipedia based on entities. Lin et al. (Lin et al. 2024) further integrated multiple external knowledge sources, including knowledge graphs, by crawling 74,000 web documents from DBpedia and Wikipedia to extract event-type knowledge, and constructed knowledge graphs comprising 10 core event categories. Vedula et al. (Vedula and Parthasarathy 2021) developed a knowledge graph for rumor detection using a large-scale structured knowledge. Wang et al. (Wang et al. 2020) introduced a knowledge-driven model, which integrates textual, conceptual, and visual information within a unified framework to learn semantic representations for fake news detection. Guo et al. (Guo et al. 2025) used ruling articles as external resources obtained from the verification websites as their basis for judgment. Hua et al. (Hua et al. 2023) addressed data sparsity issues by

Table 5 Summary of research on knowledge enhancement architecture (T: text; I: image; V: video; G: graph; A: audio)

Methods	Ref	Year	Technology	Advantage	Limitation	Featureextraction
External Knowledge-based	Song et al. 2021c)	2023	Knowledge-enhanced adversarial learning	Strong interpretability	Knowledge update is not timely	T(NLP); I(CNN)
	Zou et al. 2023)	2023	Three-layer attention-based hierarchical interaction	High knowledge usability	Difficult data cleaning	T(BERT); I(CNN)
	Lin et al. 2024)	2024	Co-attention mechanism model	Strong ability to adapt to complex models	Easy to overfit	T(BERT); I(CNN)
	Hua et al. 2023)	2023	BERT-based contrastive learning	Strong interpretability	Information is easy to overload	T(BERT); I(CNN)
Reasoning Enhancement	Qi et al. 2024)	2024	Self-supervised generative explanation model	Strong interpretability	Interpretation relies on data statistics	T(BERT); I(CNN)
	Zong et al. 2024)	2024	Diffusion model-based opinion Evolution	Relatively good theoretical foundation	High computing cost	V(LSTM); G(SNA)
	Hetzner et al. 2025)	2025	BLIP-based image-to-text generation	End-to-end training	Weak ability in specific fields	T(LSTM); I(LSTM)
	Kakizaki et al. 2025)	2025	Image-to-text generation	Wide application scenarios	Easy to lose details	ALL(LLM)
	Zheng et al. 2025)	2025	LLM debate simulation	Diversity of viewpoints	May generate wrong viewpoints	T(CLIP); I(CLIP)
	Shen et al. 2025)	2025	ERNIE2.0 knowledge enhancement	Strong interpretability	Knowledge update is not timely	T(ViT); I(ERNIE)
	Hu et al. 2024b)	2024	LLM-generated article summary	Comprehensive content summarization	Details are easy to lose	T(BERT)
	Nan et al. 2024)	2024	LLM comment simulation	Diversity of viewpoints	Details are easy to lose	ALL(LLM)

Fig. 11 The framework of external knowledge-based methods in multi-modal fake news detection. External knowledge encompasses knowledge graphs, Wikipedia, LLMs, web APIs, etc. External knowledge can be leveraged to support multi-modal fake news detection through various methods, including retrieval, representation, fusion, reasoning, verification, and alignment



back-translating news headlines and summaries from training sets using the Baidu translation API to get data amplification.

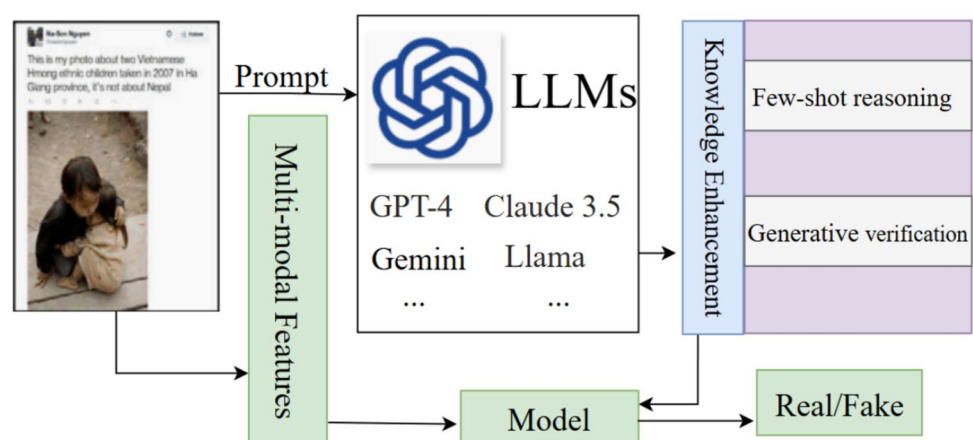
3.4.3 LLM reasoning enhancement

The emergence of LLMs presents new opportunities for detecting multimodal false news. First, multi-modal feature extraction can also be utilized by LLM pre-training models, such as GPT (Kakizaki et al. 2025) and GPT-4 V (Nan et al. 2024), among others. Through learning from large-scale data, these pre-trained models have acquired rich semantic representation capabilities that can be directly applied

to feature extraction tasks or adapted to specific fake news detection scenarios through fine-tuning techniques (Li et al. 2024c). Second, LLMs have significantly advanced multi-modal fake news detection by facilitating feature extraction, reasoning, contextual analysis, and multi-modal verification (Hasan et al. 2023; Naveed et al. 2025), as shown in Fig. 12. This mainly includes few-shot reasoning and generative verification.

Few-shot reasoning Few-shot reasoning leverages built-in knowledge base and powerful reasoning capabilities of the LLMs, achieving logical reasoning and content verification through prompt engineering (Shen et al. 2025; Dong

Fig. 12 LLM reasoning enhancement in multi-modal fake news detection



et al. 2024), entity relationship analysis (Qi et al. 2024), and logical contradiction detection (Zheng et al. 2025; Nan et al. 2024; Liu et al. 2024b). The significant advantage of this approach lies in its strong generalization capability and rapid adaptation ability, enabling it to handle novel types of fake news not seen in the training data. Cross-modal semantic consistency verification analyzes the semantic correspondence between text descriptions and image content. For example, (Qi et al. 2024) fine-tuned the Q-Former module of InstructBLIP to align general visual concepts with news domain entity recognition, identifying mismatches between headline and image information. Entity relationship analysis focuses on identifying logical associations between entities through the built-in knowledge base of LLMs. (Shen et al. 2025) utilized the ERNIE2.0 knowledge-enhanced model for deep feature extraction of text data, achieving precise analysis of entity relationships. Logical contradiction detection guides LLMs to simulate human reasoning processes. For example, (Zheng et al. 2025) guided LLMs to simulate human debate for adversarial reasoning. Similarly, (Hu et al. 2024b) leveraged GPT-3.5's reasoning capabilities to generate article summaries based on subtle clues in news content to judge authenticity, and (Nan et al. 2024) simulated online comments to let LLMs generate diverse reader comments. To facilitate adversarial reasoning, (Zheng et al. 2025) utilized the advanced capabilities of MLLMs in cross-modal understanding and text generation, which was achieved by engaging them in a debate between opposing perspectives.

Generative verification Generative verification guides LLMs to play the professional role of fact-checker through prompt engineering (Yi et al. 2025). LLMs can simulate the workflow of professional fact-checkers through structured prompt templates to conduct in-depth analysis and verification of input multi-modal content (Wu et al. 2024). In specific implementations, generative verifiers primarily utilize generative technologies, such as image-to-text generation (Kakizaki et al. 2025; LekshmiAmmal and Madasamy 2025b), to convert image content into structured text descriptions. These descriptions are then combined with original text information for comprehensive analysis and judgment (Naveed et al. 2025). (Yi et al. 2025) utilized GPT to strengthen cross-modal semantic consistency verification through generative technology, enabling the model to identify semantic correspondence between text and images. (LekshmiAmmal and Madasamy 2025b) adopted the Gemini API to generate descriptive information from images, combining Siamese network architecture and cross-modal alignment mechanisms to calculate similarity. Similarly, (Kakizaki et al. 2025) integrated image-to-text generation technology with external knowledge retrieval systems, verifying news content authenticity through external verification calls such

as the Google API. Some models utilize LLMs to rewrite news in training data into different styles, constructing a unified multi-task learning framework by integrating style-independent training and content-focused authenticity attribution prediction. For example, Wu et al. 2024) proposed the SheepDog model, introducing significant advances in generative verifiers regarding style robustness. Addressing the timeliness limitations and insufficient coverage of static knowledge (Gao et al. 2023), researchers have further explored dynamic knowledge enhancement methods (Fan et al. 2024).

4 Datasets description

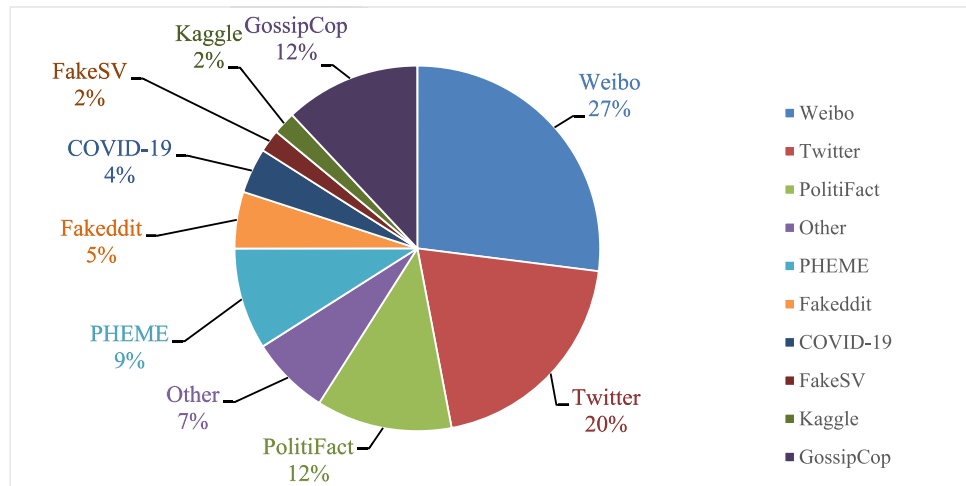
This section summarizes the commonly used datasets for multi-modal fake news detection. Among them, the proportion of data sets that use more data sets in the statistical research of this paper is analyzed, as shown in Fig. 13. The proportion from Weibo is the highest, reaching 27%, indicating that Weibo is a relatively primary information source in this field. In addition, there are Twitter, PolitiFact, and GossipCop, all with a more than 10% proportion. The detailed statistics of the data set are listed in Table 6. The following chapters describe the most commonly used data sets in recent years.

4.1 Weibo

The Weibo dataset (Jin et al. 2017), comprising 2,029 news articles about the coronavirus published between January and May 2020, along with 140,820 tweets reflecting the dissemination of this news on Twitter, has been collected. This repository contains text, visual, temporal, and network information. The dataset is stored in a specific folder and consists of two CSV files: recovery-news-data.csv and recovery-social-media-data.csv. The first file contains all news article samples collected from 22 reliable websites and 38 unreliable websites, including details such as the unique identifier of the article, URL, publisher, publication date, author, title, header image, body content, political bias, country of origin, and reliability label. The second file provides samples of the news articles' presence on social media, including unique identifiers for both the news and the corresponding tweets. Each file is comma-separated and includes relevant columns to record detailed information.

4.2 FakeNewsNet

The FakeNewsNet dataset (Nakamura et al. 2019) is multicomprising a large scale of over one million samples. It is meticulously categorized across multiple dimensions,

Fig. 13 Proportion chart of commonly used datasets**Table 6** Multi-modal datasets for fake news detection (T: text; I: image; N: network; V: video)

DataSet	Year	T	I	N	V	classes	Data link
Weibo	2017	✓	✓	✓		2 classes: fake/real	https://gitcode.com/open-source-toolkit/53b01
Twitter	2015	✓	✓	✓		2 classes: fake/real	https://github.com/MKLab-ITI/image-verification-corpus
PHEME	2017	✓	✓			2 classes: fake/real	https://figshare.com/articles/PHEME_dataset_of_rumours_and_non-rumours/4010619
PolitiFact	2018	✓	✓	✓		2 classes: fake/real	https://github.com/KaiDMML/FakeNewsNet
GossipCop	2018	✓	✓	✓		2 classes: fake/real	https://paperswithcode.com/dataset/fakenewsnet
Fakeddit	2019	✓	✓			7 classes: False, ironic, misleading, false association, etc	https://github.com/entitize/fakeddit
Buzzfeed	2021	✓	✓		✓	2 classes: fake/real	https://github.com/BuzzFeedNews/2016-10-facebook-fact-check
FakeSv	2023	✓	✓		✓	4 classes: fake/real/debunked/irrelevant	https://github.com/ICTMCG/FakeSV
MuMiN	2022	✓	✓	✓		2 classes: fake/real	https://github.com/shivangi-aneja/COSMOS
COSMOS	2021	✓	✓			2 classes: fake/real	https://github.com/shivangi-aneja/COSMOS
MM-OVID	2021	✓	✓	✓		2 classes: fake/real	https://openxlab.org.cn/datasets/OpenDataLab/MM-COVID
CHECKED	2021	✓	✓	✓		2 classes: fake/real	https://github.com/cyang03/CHECKED
CoAID	2020	✓	✓		✓	2 classes: fake/real	https://github.com/cuilimeng/CoAID
MC-Fake	2021	✓	✓	✓		2 classes: fake/real	https://www.nactem.ac.uk/mc-fake/
FineFake	2024	✓	✓	✓		2 classes: fake/real	https://github.com/TrustworthyComp
MCFEND	2024	✓	✓			2 classes: fake/real	https://github.com/TrustworthyComp
MR2	2022	✓	✓	✓		3 classes: rumor/non-rumor/unverified	https://github.com/THU-BPM/MR2
ReCOVery	2023	✓	✓	✓		2 classes: fake/real	https://github.com/apurvamulay/ReCOVery

including news text, image data, metadata, and comment data. Specifically, the dataset is rich in content, encompassing the news body, associated images, and reader comment data. The dataset includes subsets of PolitiFact and GossipCop.

The PolitiFact dataset includes 1,152 news articles crawled from the political fact-checking website PolitiFact, labeled as "true" or "false". It collects celebrity gossip and

entertainment news from May 2002 to July 2018. It includes labeled news text, engagement content, and social content. The news text comprises the id, title, text content, URL, and news author. Engagement encompasses user interactions with the news, while social content reflects users' social connections on social media.

The GossipCop subset includes 22,417 pieces of entertainment news collected from the entertainment news

verification platform GossipCop. It collects news about politics from July 2000 to December 2018, which humans have manually labeled as true or false. The engagement content includes user interactions with the news, while the social content reflects users' social connections on social media.

4.3 Twitter

The dataset comprises multiple data files, containing nearly 20,000 tweet entries, including those from MediaEval 2015 (Boididou et al. 2015) and MediaEval 2016 (Larson et al. 2017). The Mediaeval 2015 includes 2,433 news items that have been verified as either fake or real. This dataset provides precise details, including tweet_id, tweet text, user_id, image_id, username, tweet timestamp, and a classification indicating whether the content is real or fake. Specifically, it includes images_txt, which contains image data that has been verified as real or fake. The dataset includes the image_id field for each image, the image_url field for the image's URL, annotations confirming the image's authenticity, and the event from which the image originates. tweets_images contain tweet data used to form the dataset. This file includes the tweet_id field displaying the id of each tweet, the image_id field for the associated image, annotations marking the authenticity of each tweet, and the event from which the tweet originates.

4.4 Pheme

The Pheme dataset (Zubiaga et al. 2017) is based on 5 breaking news events and provides fake and real judgments for tweets related to these events. The dataset contains a total of 1,972 fake tweets and 3,830 real tweets. For each of the 5 Healthcare Misinformation folders, labeled fake and non-fake. These two folders contain subfolders named after the tweet IDs. The tweets themselves can be found in the source-tweet directory of the corresponding tweet, while the 'reactions' directory contains the tweets responding to the source tweet.

4.5 Fakeddit

The Fakeddit dataset (Nakamura et al. 2019) is multi-modal, extensively covering samples from multiple news categories with a vast scale, exceeding millions of samples. Specifically, this dataset is rich in content, encompassing news articles, images associated with the news, and reader comment data.

4.6 Buzzfeed

The BuzzFeed news dataset (Shu et al. 2020) includes news posted by nine news organizations on Facebook during the week before and after the 2016 U.S. presidential election. It contains 91 fake news and 91 real news articles, all saved

in csv format. Each record includes 12 feature variables, such as the unique id of the news webpage, the headline, the body text, the author, and publisher information, as well as links to images and videos in the body. BuzzFeed journalists have rigorously verified the dataset to ensure its authenticity.

4.7 FakeSV

The FakeSV dataset (Qi et al. 2023) crawled 99,970 rumors from January 2011 to January 2022. 854 event descriptions were finally sorted out as search queries. Relevant video content, user replies, and publisher information are retrieved from the chatter and fast hand crawls. The dataset contains video, cover image, title, and publication time. The dataset contains 1,827 fake news videos, 1,827 real news videos, 1,884 exposed videos, and 738 events.

4.8 COVID-19

The CoAID-19 dataset (Cui and Lee 2020) presents the CoAID Covid-19 healthcare misinformation dataset, with diverse COVID-19 healthcare misinformation, including fake news on websites and social platforms and users' social engagement about such news. CoAID includes 4,251 news articles, 296,000 related user engagements, 926 social platform posts about COVID-19, and ground truth labels.

4.9 Kaggle

The Kaggle dataset, also known as the ISOT Fake News detection dataset for binary text classification, is divided into two files: "Fake.csv", which contains 23,502 fake news articles, and "True.csv", which has 21,417 real news articles. Each article in the dataset includes columns such as title, body text, subject, and publish date. The datasets used in this research are open benchmark datasets provided by the portal Kaggle (Szczepański et al. 2021).

4.10 RumourEval

The RumourEval dataset (Gorrell et al. 2018) is a shared dataset that addresses two core NLP tasks: veracity prediction and stance detection. It mainly supports relevant tasks in the SemEval International Workshop on semantic evaluation. The dataset primarily consists of 297 source rumour tweets and 7,100 discussion tweets. Most source rumours are derived from English-language platforms such as Snopes.

4.11 MediaEval

This MediaEval dataset, a component of the 2016 MediaEval task, is publicly accessible via the Image

Verification corpus MediaEval 2016. It comprises tweets related to multiple events and has been widely utilized in research, owing to Twitter's status as one of the most commonly used social media platforms and its high susceptibility to creating and propagating fake news. The dataset includes attributes such as post_id, post_text, user_id, image_ids, username, timestamp, and label, with three key attributes-post_text the textual content of the tweet, image_ids a link to the tweet-associated image, and label indicating authenticity as real or fake selected for use, the dataset contains 15,519 image-associated tweets with binary labels, including 6,875 fake tweets and 8,644 real tweets.

4.12 DGM4

The DGM⁴ dataset (Shao et al. 2023) comprises 230,000 news samples, including 77,426 pristine image-text pairs and 152,574 manipulated pairs. The manipulated pairs comprise 66,722 face swap manipulations, 56,411 face attribute manipulations, 43,546 text swap manipulations, and 18,588. Most manipulated images have small regions of manipulation, and the manipulated text tokens are relatively few. It makes the DGM4 dataset more challenging for forgery detection than existing deepfake and multi-modal disinformation datasets.

4.13 MR²

The MR² dataset (Li et al. 2024c) comprises two versions: English MR2-E and Chinese MR2-C, originating from the Twitter and Weibo platforms, respectively. It contains 14,700 real-world posts, of which MR2-C has 1,754 rumors, 2,609 non-rumors, 3,361 unverified posts, and 497,233 threads. MR2-E has 1,418 rumors, 2,318 non-rumors, 3,240 unverified posts, and 576,254 threads, all of which

have been removed from pure text posts and subjected to data quality control such as deduplication and filtering. The training, development, and testing sets are divided in an 8:1:1 ratio.

5 Experimental discussion

This section discusses commonly used fake news detection methods, compares their results on popular datasets, and conducts statistical analysis.

5.1 Evaluation matrix

Multi-modal fake news detection is a binary classification problem that aims to classify input news samples, which contain both textual and visual information, as either real or fake. This task requires models to comprehensively analyze textual content, visual information, and the semantic consistency between them to make accurate authenticity judgments.

Model performance evaluation employs multiple standard classification metrics for this binary classification task. The detailed definitions and calculation formulas for the evaluation metrics are organized in a tabular form, with specific content presented in Table 7.

5.2 Experimental comparison

Comparative experimental results of different methods on the same datasets show significant performance differences between models, as shown in Table 8. For the Weibo and Twitter datasets, the EMAF and SGAMF models demonstrate obvious advantages in prediction results, sharing a common characteristic: they are both

Table 7 Multiple standard classification metrics for model evaluation

Assessment criteria	Brief description
$\text{accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$	Accuracy refers to the ratio of correctly predicted samples to the total number of samples (Qi et al. 2024). It is an intuitive metric representing the model's overall ability to predict correctly. However, it is not sensitive enough for imbalanced datasets, and when there is a significant imbalance between positive and negative samples, a high accuracy may mask the model's insufficient prediction of the minority class
$\text{Precision} = \frac{TP}{TP+FP}$	Precision refers to the proportion of actual positive samples among the samples predicted by the model. High precision means that a larger proportion of the samples predicted as positive are positive, with fewer false positives
$\text{Recall} = \frac{TP}{TP+FN}$	Recall refers to the proportion of actual positive samples correctly predicted as positive by the model. It reflects the model's ability to identify positive samples. A high recall means the model can correctly identify more positive samples with fewer false negatives
$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	The F ₁ score is the harmonic mean of Precision and Recall, used to evaluate the overall performance of a model. The F1 score balances Precision and Recall, making it a comprehensive metric. It is particularly suitable for situations with imbalanced positive and negative samples, as it simultaneously considers the model's reliability and identification ability

entity-based detection methods. It clearly shows that this entity alignment approach, which constructs cross-modal associations at the semantic level for semantic-level understanding models, is highly efficient in fake news detection.

CreditRank and DPSG are the top-performing models on the PolitiFact and GossipCop datasets. Both approaches build models from the perspective of social context modeling by analyzing the information dissemination process on platforms. Both utilize graph-based interactions as their core

feature source, leveraging entity behavior to compensate for the shortcomings of content semantics, significantly improving detection robustness.

Accuracy and other benchmark metrics demonstrate that high-performing models are often designed based on entities. Their behavioral trajectories and relationship networks are robust and difficult to tamper with, providing an independent verification dimension for content authenticity. Fake news is essentially a combination of false content and anomalous dissemination, and entities precisely cover both

Table 8 Comparison of experimental results using different methods

Datasets	Papers	Methods	Accuracy	Precision	Recall	F ₁
Weibo	EMAF(Li et al. 2021)	Entity matching mechanisms	0.974	0.954	0.998	0.976
	TGNF(Song et al. 2021b)	Temporal memory module	0.968	0.962	0.975	0.969
	DA-GCN(Liu et al. 2022)	Content-graph dual attention mechanism	0.944	0.941	0.946	0.944
	CLKD-IMRD(Xu et al. 2023b)	Self-attention	0.933	0.933	0.929	0.931
	MACCN(Yi et al. 2024)	Adding synonym replacement	0.924	0.921	0.872	0.901
	MMFN(Yang et al. 2021)	Cross-model fusion	0.923	0.921	0.926	0.924
	HCCIN(Wu et al. 2023c)	Cognition-inspired collaborative reasoning	0.915	0.921	0.973	0.946
	TRIMOON(Xiong et al. 2023)	Gate-based fusion module	0.912	0.929	0.888	0.908
	MMCAN(Hu et al. 2024a)	Matching-aware cooperative attention	0.911	0.913	0.91	0.912
	1D-CCNet(Yan et al. 2024)	One-dimensional attention	0.909	0.838	0.963	0.894
	MRHFR(Wu et al. 2023a)	Reading habit-based consistency reasoning	0.907	0.939	0.869	0.903
	CARMN(Song et al. 2021a)	Trans-modal attention mechanism	0.825	0.826	0.825	0.825
	EANN(Wang et al. 2018)	Adversarial networks	0.816	0.806	0.795	0.8
	Singh et al.(Singh and Sharma 2022)	Self Attention-Transformer	0.812	0.851	0.784	0.816
	MFN(Chen et al. 2021)	Self-attention mechanism	0.808	0.808	0.806	0.807
	att-RNN(Jin et al. 2017)	CNN of attention mechanism	0.788	0.862	0.686	0.764
Twitter	SGAMF(Du et al. 2024)	Sparse gated attention mechanism	0.950	—	—	0.940
	MMCAN(Hu et al. 2024a)	Matching-aware cooperative attention	0.943	0.869	0.976	0.919
	MMFN(Yang et al. 2021)	Cross-model fusion	0.935	0.960	0.856	0.905
	1D-CCNet(Yan et al. 2024)	One-dimensional attention	0.925	0.922	0.924	0.922
	CLKD-IMRD(Xu et al. 2023b)	Self-attention	0.880	0.881	0.846	0.865
	Singh et al.(Singh and Sharma 2022)	Self Attention-Transformer	0.853	0.821	0.943	0.877
	MFN(Chen et al. 2021)	Self-attention mechanism	0.806	0.799	0.777	0.785
	EMAF(Li et al. 2021)	Entity matching mechanisms	0.805	0.883	0.756	0.815
	EANN(Wang et al. 2018)	Adversarial networks	0.788	0.778	0.768	0.787
	att-RNN(Jin et al. 2017)	CNN of attention mechanism	0.682	0.78	0.615	0.689
PolitiFact	CreditRank(Jarrahi and Safari 2023)	User credibility assessment integration	0.99	0.979	1.000	0.989
	DPSG(Jing et al. 2025)	Temporal fusion	0.985	1.000	0.971	0.986
	TieFake(Guo et al. 2023)	Similar points accumulate attention	0.912	0.931	0.909	0.92
	SceneFND(Yi et al. 2025)	Processing specific types of entities	0.832	0.8479	0.8491	0.832
	EANN(Wang et al. 2018)	Adversarial networks	0.795	0.813	0.761	0.786
GossipCop	DPSG(Jing et al. 2025)	Temporal fusion	0.991	0.979	0.980	0.98
	CreditRank(Jarrahi and Safari 2023)	User credibility assessment integration	0.988	0.985	0.966	0.975
	TieFake(Guo et al. 2023)	Similar points accumulate attention	0.892	0.887	0.902	0.894
	ARG(Hu et al. 2024b)	LLM Reasoning Enhancement	0.878	—	—	0.79
	EANN(Wang et al. 2018)	Adversarial networks	0.833	0.842	0.835	0.838
	SceneFND(Yi et al. 2025)	Processing specific types of entities	0.748	0.7487	0.7531	0.7471

dimensions. These core advantages enable entity-based detection methods, such as EMAF, SGAMF, CreditRank, and DPSG, to achieve optimal performance across diverse datasets and scenarios, making them a key technical direction in the field of fake news detection. In contrast, poorly performing models rely solely on traditional methods such as at-RNN, which have inherent limitations in multimodal tasks—they remain at the traditional feature-level fusion level, which is particularly evident in text-image fusion scenarios.

These core advantages enable entity-based detection models, such as EMAF, SGAMF, CreditRank, and DPSG, to achieve optimal performance across various datasets and scenarios, making them a key technical direction in the field of fake news detection.

6 Discussion

This section summarizes the advantages and disadvantages of typical classification methods, which are shown in Table 9. Researchers can select the appropriate model for reference research based on their application needs.

7 Conclusion and future direction

This paper reviews multi-modal fake news detection research progress and summarizes it from a DL framework and technology perspective. First, we introduce the original classical modal interaction methods based on fusion approaches. Secondly, the attention mechanism, GCN, and Transformer, among others, have improved detection accuracy. Meanwhile, the performance of fake news detection has been improved with the vigorous development of entity-based methods, social network-based methods, knowledge enhancement and LLMs reasoning methods, etc. It is a wise choice for researchers to use a variety of fusion models to combine the characteristics of different modalities and make a comprehensive classification. Through the in-depth study of fake news detection, although the existing research has made practical efforts, there are still many challenges.

1. Dataset challenges.

- **Acquisition and annotation.** New news data continues to emerge, making it challenging to obtain fake news. There is an imbalance in the number of true and false categories in multi-modal news data, and standard data requires

professional knowledge and a large amount of manpower and physics, resulting in high annotation costs.

- **The classification of dataset granularity is coarse.** Most studies use news categorized into two types: real and fake. It is challenging to handle mixed fake news data that may originate from diverse sources or exhibit inconsistent timestamps.

2. Cross-modal semantic understanding.

- **Semantic consistency.** In multi-modal data, the semantic consistency of different models is a key problem. A key research direction is to effectively integrate different information models to enhance the accuracy of fake news detection.
- **Data heterogeneity.** Multi-modal data of different modes may be inconsistent in time, space, and semantics. So, the alignment between text, image, and video is a key problem for multimodal data.

3. Specific features and scenarios.

- **Dynamic characteristics of news.** With the emergence of new news, news characteristics may change over time. The existing detection models have difficulties in identifying fake news early, which has not been well addressed.
- **Cross-language and cross-cultural research.** The social media audience comprises individuals from diverse cultural, age, and educational backgrounds. Significant differences exist in word usage, emotional expression, and writing styles across different languages and cultural regions. These differences often lead to performance degradation in cross-domain detection models. It is crucial to address the challenge of curbing the global spread of misinformation and enhancing the international applicability of methods for detecting fake news.
- **Deep fake of multi-modal data.** While some methods for processing multi-modal news can detect image tampering, deepfake detection for video and audio still faces significant technical bottlenecks. Furthermore, detection models often lack transparency in revealing their tampering logic.

4. Optimization of the detection model

- **The challenge of the LLMs.** The emergence of LLMs makes it easier to generate and rewrite fake news, increasing the difficulty of identifying misinformation through text-based methods. Additionally, LLMs address challenges such as inconsistent graph-

Table 9 Analysis of advantages and disadvantages, and guidance on selection for existing methods

Methods	Model	Advantages	Disadvantages
General content-based fusion	att-RNN(Jin et al. 2017)	One of the most classic articles in the first generation of using attention mechanisms for multimodal fake news detection is commonly used for comparison, as it achieves a deep fusion of the "text-image-social context" three modalities and solves cross-modal feature alignment problems through attention mechanisms	Limited by traditional methods, the model's attention mechanism only generates attention vectors based on text-social context LSTM outputs, only acting on single-granularity matching of text word-level-LSTM hidden states and visual neuron-level, without considering semantic hierarchical differences between modalities
	CARMN(Song et al. 2021a)	Combining a multi-modal attention residual network and a multi-channel CNN to integrate data more effectively	High computational complexity
	MPFN (Jing et al. 2023)	Effectively captures the representation information of each mode at different levels, avoiding the problem that existing methods only rely on deep features	The processing and fusion of multimodal data may lead to information loss, especially in cases of data scarcity or imbalance
Cross-modality Transformers	SV-FEND(Qi et al. 2023)	Combining five modalities, including comments and users for processing, adopting two dedicated cross-modal Transformer layers, taking comments, user feedback and users as core inputs for two types of social context features, fusing with news content features through a self-attention Transformer, utilizing social signals to assist detection, compensating for the limitations of relying solely on content	The authors note in ablation experiments that after removing comment features, model accuracy drops from 79.31% to 79.09%, with a minimal difference, indicating that SV-FEND has low utilization efficiency for comment features
	AKA-Fake(Zhang et al. 2024b)	Key parameters support linear adjustment, which can reduce clustering layers by lowering ρ , resulting in a 30% reduction in computational time at the cost of sacrificing 1–2 percentage points of F1 value	If the target domain KG's relationship types differ significantly from the general KG's, meta-paths must be redesigned
	MRE-FND(Han et al. 2024)	Graph sampling reduces redundant computation, alleviating the phenomenon of graph node neighbor quantities growing exponentially with sampling order, causing computational resource overload and a sharp decline in training efficiency	Limited expansion capability for unstructured data
Propagation network	SureFact (Yang et al. 2022)	It removes irrelevant information through subgraph reasoning to enhance the model's generalization and differentiation abilities	It is not easy to cope with large-scale news networks
	MVFN(Cui et al. 2023)	By constructing visual-text cross-modal graphs and social context graphs, and using Graph Attention Networks (GAT) to mine latent semantic connections in different modalities and social relationships, a modality alignment mechanism is introduced to make features from different views more consistent, facilitating better fusion	Low interpretability, and the model needs to construct two graphs (visual-text cross-modal graph and social context graph) and perform feature aggregation via GAT. Although the effect is good, it increases computational resource costs, making this approach less cost-effective
Temporal-based fusion	TGNF (Song et al. 2021b)	Considering timing, it can handle real-time data flow and network changes	Insufficient generalization ability, unable to efficiently handle massive data
	DPSG(Jing et al. 2025)	Through "temporal dynamic fusion+deep graph structure mining+Transformer global perception," it addresses the core problems of traditional temporal schemes: "static fusion, shallow structure, and long-sequence forgetting."	The dynamic fusion mechanism has a strong dependence on the integrity of data timing. Dynamic fusion easily degrades into static concatenation if time records are incomplete or the news has no propagation sequence

Table 9 (continued)

Methods	Model	Advantages	Disadvantages
External knowledge-based	TTEC (Hua et al. 2023)	It still has a strong ability to distinguish between complex news sources and content	Some fine-grained vacation news may need more information sources, but current methods may not fully capture all features
	MFEK (Lin et al. 2024)	It integrates explicit and implicit knowledge, using Wikipedia to obtain explicit knowledge and LLMs such as ChatGPT to extract implicit knowledge	External knowledge is rich but inevitably introduces more noise, leading to a slight drop in model performance
Cross-modal LLM reasoning enhancement	SNIFFER (Qi et al. 2024)	It finds the inconsistency in pairing images and texts and analyzes the relationship between statements and evidence	The data quality may limit the model's accuracy when dealing with highly complex or blurred image and text pairing
	ARG (Hu et al. 2024b)	The reason-free version constructed via knowledge distillation avoids querying the LLM for every prediction, reduces computational costs, and is more suitable for cost-sensitive scenarios	Data modalities are relatively single, and the approach focuses only on LLMs' reasoning; there is room for improvement in subsequent inter-modal interactions

ics and text alignment, as well as their currently underdeveloped multimodal performance.

- **The interpretability of the model needs to be explored.** The model's decision-making process is opaque, making it difficult for users to trust the detection results and affecting its practical application. It is necessary to develop an interpretable framework to help better understand and interpret the prediction of multi-modal detection models. Combining the reasoning ability of LLMs with the structured representation of domain knowledge and constructing a dynamic, traceable interpretation framework will be key to improving the practicality of the model.

Unfortunately, as research on fake news detection is a rapidly evolving field, numerous models and research methods have emerged in recent years and are continually updated, making it challenging to identify all the available methods. It remains a topic worth exploring and summarizing in the future to draw researchers' attention to new methods and theories of fake news detection.

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